



Creating a local AI and building a
web UI

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Purpose:

The purpose of this lab was to learn how to build an AI that locally runs on our personal network. We also needed to learn how to build a UI for the AI through Python and/or docker, which either required some basic python knowledge or some basic Linux terminal knowledge.

Background:

We needed to get a local AI up and work on our PCs for the first AI. Ollama and LM studio both make local AI used only on personal computers. Ollama has the benefit of being simpler than LM studio, but LM studio is better because of its customizability. We decided to go with ollama because it has a UI that works on Linux but not windows.

Ollama allows you to choose an AI that is open source and download it onto your personal computer. Ollama has extensive models to choose from and install onto your computer, which all have different parameter counts. You can then run the AI locally on your computer in the command line. While Ollama does allow you to run the ai on your own computer rather than using public servers, you also need a powerful system that will allow you to run that AI. You will also have to use a lower parameter model that may not yield as good of answers if a difficult question is asked. While this could affect your accuracy, it usually will not affect your answer on more basic everyday tasks, and higher parameter counts will just slow the response down. At big AI companies like OpenAI they use trillions of parameters to run their ai, but they have massive servers that run with very large amounts of vram that are not affordable for anyone.

A benefit of running AI locally rather than using public servers is that the data that you enter the ai is private and cannot be tracked or sold. Many AI companies can use that data that you enter the AI and sell it to other companies. Many companies want and need to use local AI like what we have setup because Employees will frequently enter confidential information into AI releasing this information into the AI and to the large AI companies that will turn around and sell the data.

While Ollama does allow you to privately run an AI, it does not natively support an UI on windows, which is the operating system that most people

use. To set up a UI we used open WebUI and put it in a docker container so that we can run it in Linux. Docker functions similar to a virtual machine. A virtual machine uses resources on your own computer to virtually run an entire separate operating system on your own computer. This requires decent hardware on your computer, which has more RAM than most. A virtual machine also takes much longer to run than a docker container. A Docker container isolates an operating system and runs a slightly striped down version of that chosen operating system. If you have an app that does not natively support the operating system that you are using, you can sometimes use a docker container and put the app inside of the container. Using a virtual machine would use more resources than you would need to use and would run much slower.

A VM is running on hardware and virtualizes the hardware that it is running on. This allows you to use a virtual operating system and things on that operating system. While it takes a bit of time to set up and get running, it allows you to run a full operating system virtually. Docker does not do the same thing. Docker allows you to virtualize the operating system rather than virtualize the hardware. This allows you to run something called a docker container with docker. A docker container is similar to how you run an operating system on virtualized hardware, but a docker container is much more lightweight and much more isolated than a virtual machine. Docker containers are so isolated that they basically don't interact at all with anything outside of the container unless told otherwise. While docker containers do come with a striped down basic version of Linux installed on them, you can very easily just pull a known Linux distro from the internet and install it into the docker container. This can be done with the docker pull command. For open WebUI to work it needs Linux installed in the container. This is one of the reasons we used docker. While a VM will work to also run Open WebUI, docker is lightweight, easy to use, and faster than a VM as well as more isolated.

Lab Summary:

We set up and successfully got an AI to run locally from our computer on the network. We were able to connect to the AI from anywhere on the same network. We also had to set up a docker with a docker container. We had to put open WebUI in the docker container as it had to be on Linux to run. We then set up Open WebUI's UI that works with ollama so that we could make a UI for the AI that had been made. We then made a UI using python that was an alternative to using Open WebUI, and I think that using

python is better than Open WebUI because you don't have to sign into open WebUI and you have further customizability.

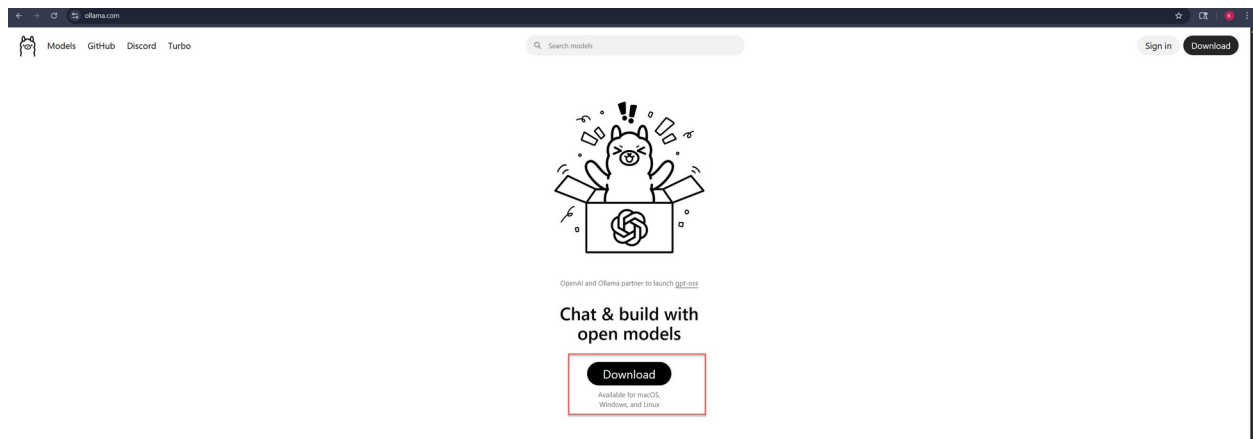
Lab Commands:

Ollama run: will both run the AI of choice and download if it is not installed already. Make sure that you specify the parameter count.

Docker run: creates the docker container you can only use this command once for the same container then use the docker exec command to enter configuration.

Dvmt: something that you can install on Linux that allows you to run multiple terminals on one terminal.

Curl: Curl is used to download and upload data on Linux



Go to [Ollama](https://ollama.com) and click the download link to download ollama.

Download Ollama



macOS



Linux

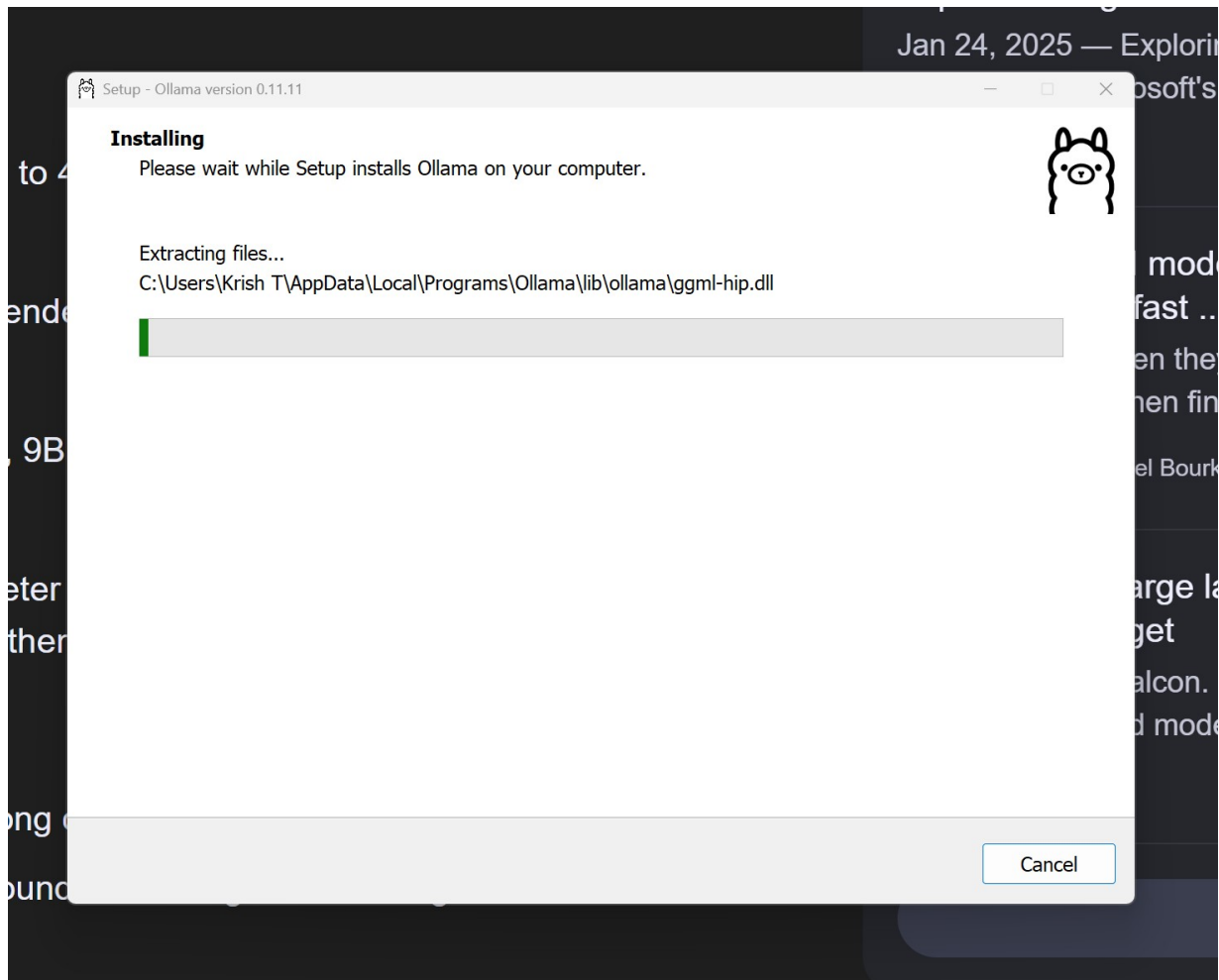


Windows

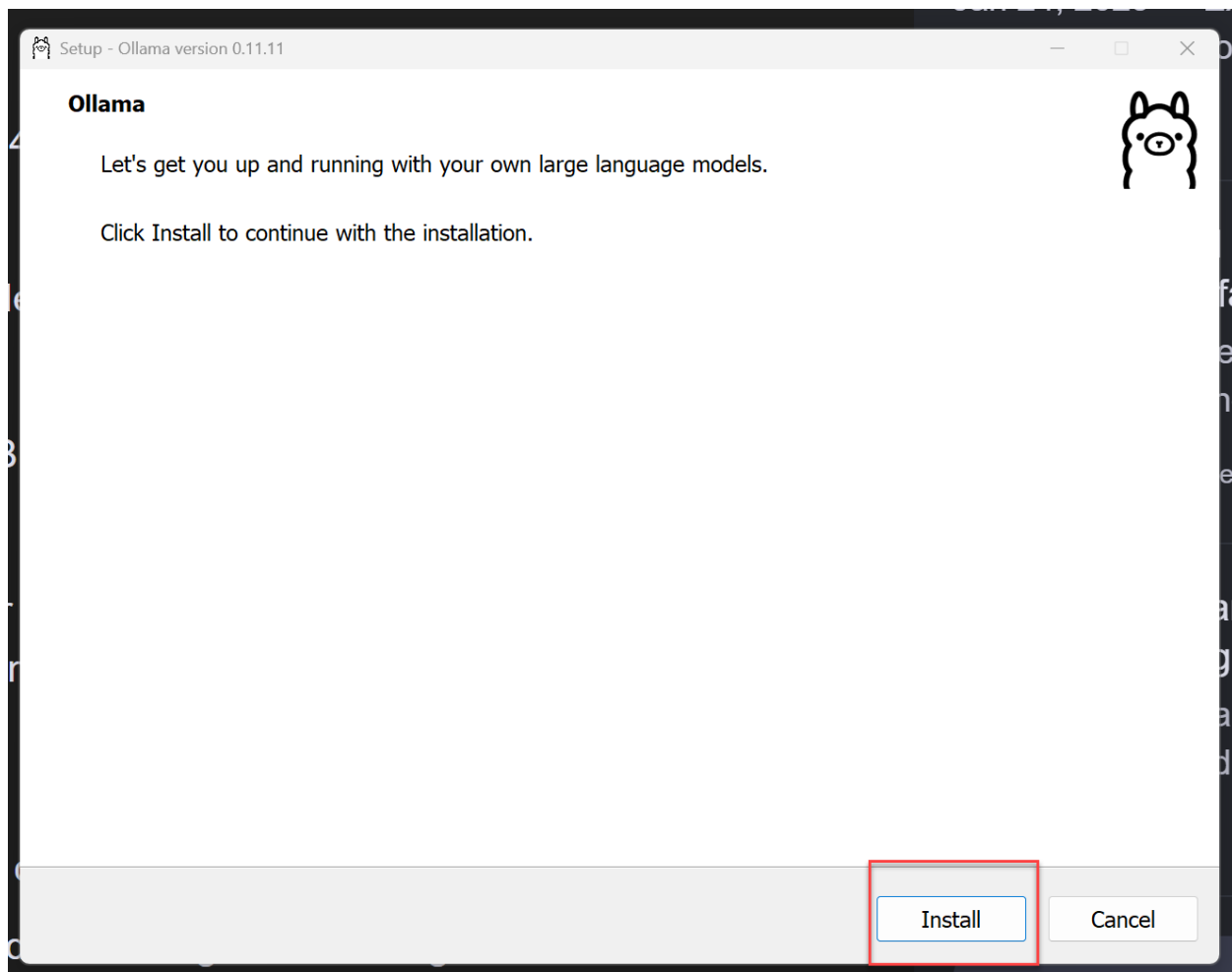
Download for Windows

Requires Windows 10 or later

Download ollama for your operating system (probably windows)



A message like this should show up showing that ollama is installing.



Once it has finished, click install.

```
Command Prompt
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Krish T>
```

To check the install go to command prompt

```
Command Prompt
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Krish T>ollama
Usage:
  ollama [flags]
  ollama [command]

Available Commands:
  serve    Start ollama
  create   Create a model
  show     Show information for a model
  run      Run a model
  stop     Stop a running model
  pull     Pull a model from a registry
  push     Push a model to a registry
  list     List models
  ps       List running models
  cp       Copy a model
  rm       Remove a model
  help     Help about any command

Flags:
  -h, --help      help for ollama
  -v, --version    Show version information

Use "ollama [command] --help" for more information about a command.

C:\Users\Krish T>
```

Type in ollama into command prompt to verify that ollama is installed.


```
Command Prompt - ollama r x + v
(c) Microsoft Corporation. All rights reserved.

C:\Users\Krish T>ollama
Usage:
  ollama [flags]
  ollama [command]

Available Commands:
  serve    Start ollama
  create   Create a model
  show     Show information for a model
  run      Run a model
  stop     Stop a running model
  pull     Pull a model from a registry
  push     Push a model to a registry
  list     List models
  ps       List running models
  cp       Copy a model
  rm       Remove a model
  help     Help about any command

Flags:
  -h, --help      help for ollama
  -v, --version    Show version information

Use "ollama [command] --help" for more information about a command.

C:\Users\Krish T>ollama run gemma3
pulling manifest
pulling aeda25e63ebd: 1% | 25 MB/3.3 GB 4.2 MB/s 13m15s
```

to install the local ai onto your computer use the "ollama run" command, and the desired open-sourced AI along with the supported number of parameters if you want to specify. For example, "ollama run llama3.1:8b". Ollama supports tons of different open sourced ai. Here is a link all ollama's supported ai: [Ollama Search](#)

```
pulling b6ae5839783f: 100% 489 B
verifying sha256 digest
writing manifest
success
>>>
Okay, I'm here! 😊

To help me assist you effectively, could you tell me:

* **What do you need help with?** Are you looking for information, need me to write something, want to brainstorm ideas, solve a problem, play a game, or something else entirely?
* **Can you give me a bit more context?** The more details you provide, the better I can understand your request and provide a relevant response.

For example, you could say:

* "I need help writing a short poem about a sunset."
* "Can you explain the concept of blockchain to me?"
* "I'm trying to come up with a name for a new coffee shop. Any ideas?"

I'm ready when you are!
>>> whats your name
That's a great question! 😊

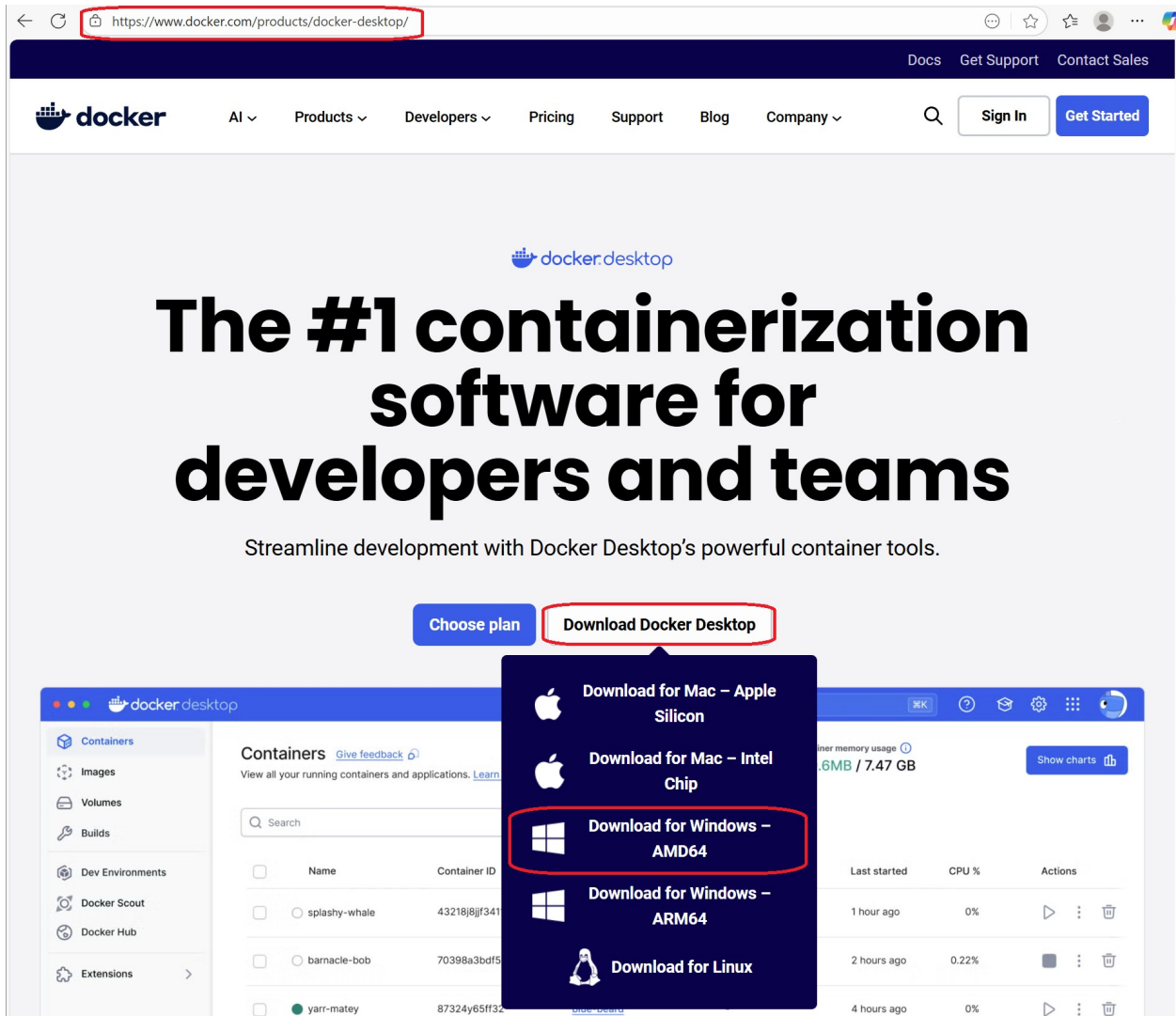
I don't really *have* a name in the way humans do. I'm a large language model, trained by Google. You can just call me Bard.

It's nice to meet you! How can I help you today?

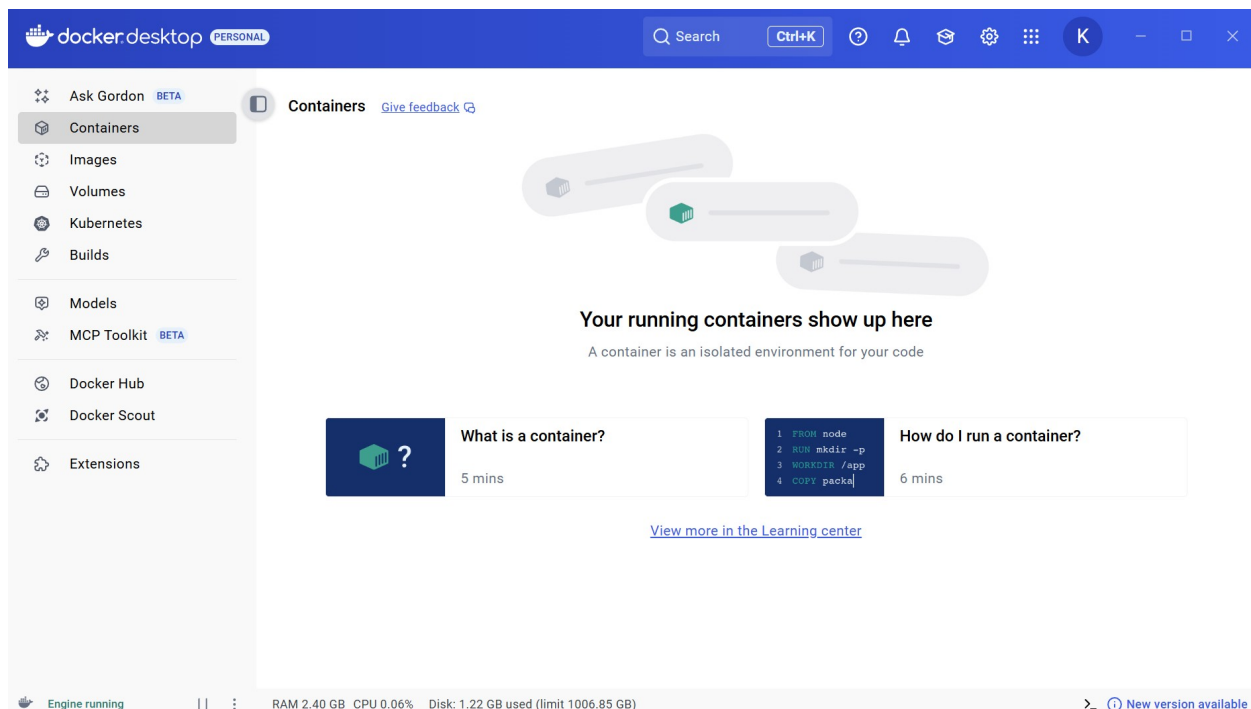
>>> /bye

C:\Users\Krish T>
```

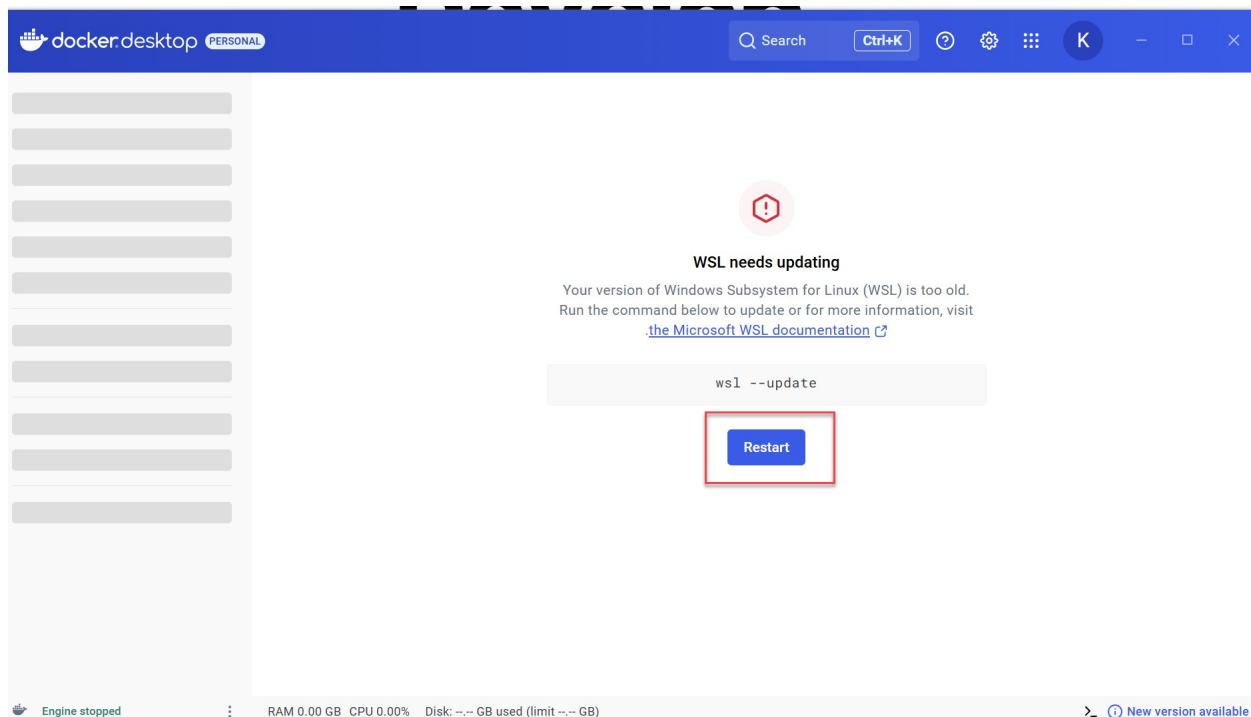
Wait for the ai model to be installed and you should be able to chat with it in the command line.



While the AI does work in the command line, we want to have a UI for the AI as well as a way to connect to it locally on our network. To do this we are going to use docker and open webUI. First go to [Docker Desktop: The #1 Containerization Tool for Developers | Docker](https://www.docker.com/products/docker-desktop/) and install docker.



Once docker is installed run the docker desktop app. Docker may require you to make an account so create one. Migrate to the containers tab on the left and side of the GUI. You should not have any containers right now, but we will make one so keep this open for later.



If you do happen to get a message like this, just click restart. This will restart your computer so make sure that you have everything saved.

```
PS C:\Users\Luke M> docker --version
Docker version 28.4.0, build d8eb465
```

Verify that docker is installed on your computer by going to PowerShell and entering the command `docker --version`.

<https://docs.openwebui.com>

WebUI

Started >

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Quick Start with Docker

INFO

WebSocket support is required for Open WebUI to function correctly. Ensure that your network configuration allows WebSocket connections.

If Ollama is on your computer, use this command:

```
docker run -d -p 3000:8080 --add-host=host.docker.internal:host-gateway -v open-we
```

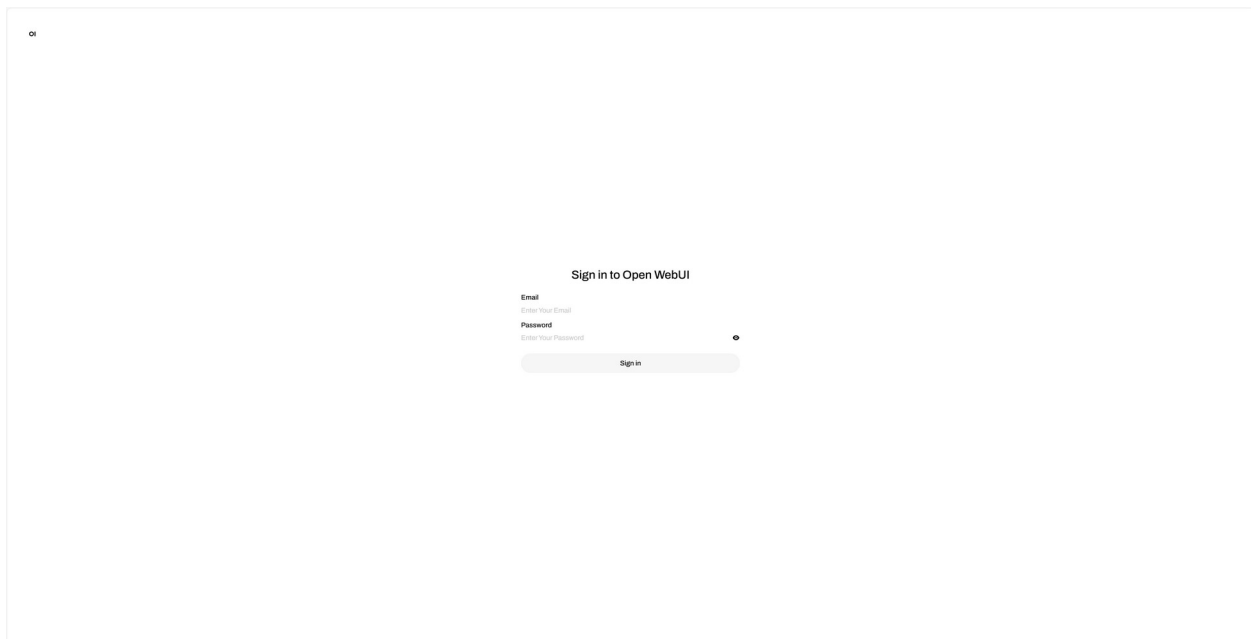
go to [Home | Open WebUI](#) and copy the first docker run command under the text "If Ollama is on your computer".

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

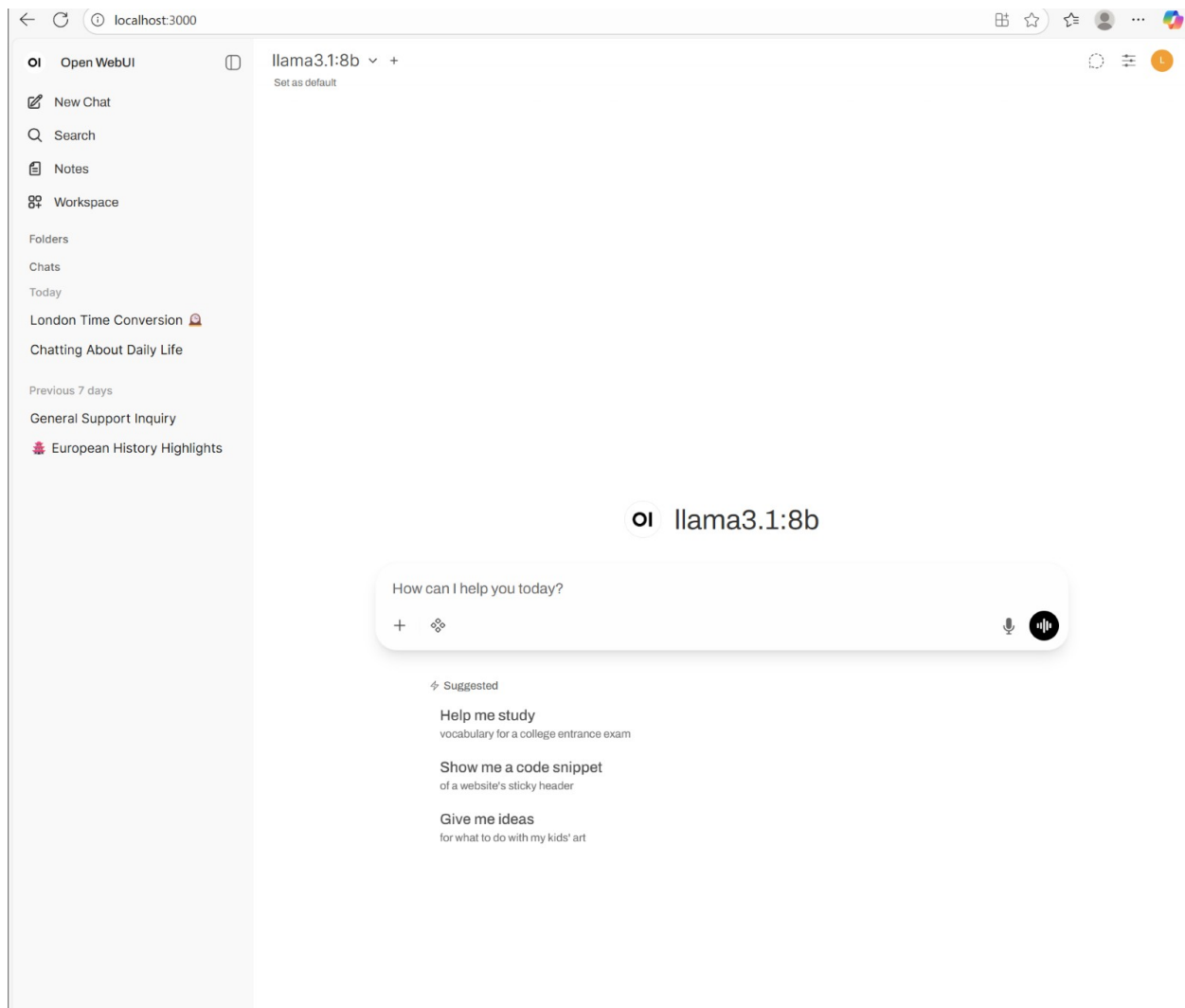
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Luke M> docker run -d -p 3000:8080 --add-host=host.docker.internal:host-gateway -v open-webui:/app/backend/data --name open-webui --restart always ghcr.io/open-webui/open-webui:main
```

Go to PowerShell and right click to paste the command into PowerShell. This will create a container in docker that isolates the apps inside the container from the rest of the OS. A minimal form of Linux was already installed in the container before the docker run command was issued.



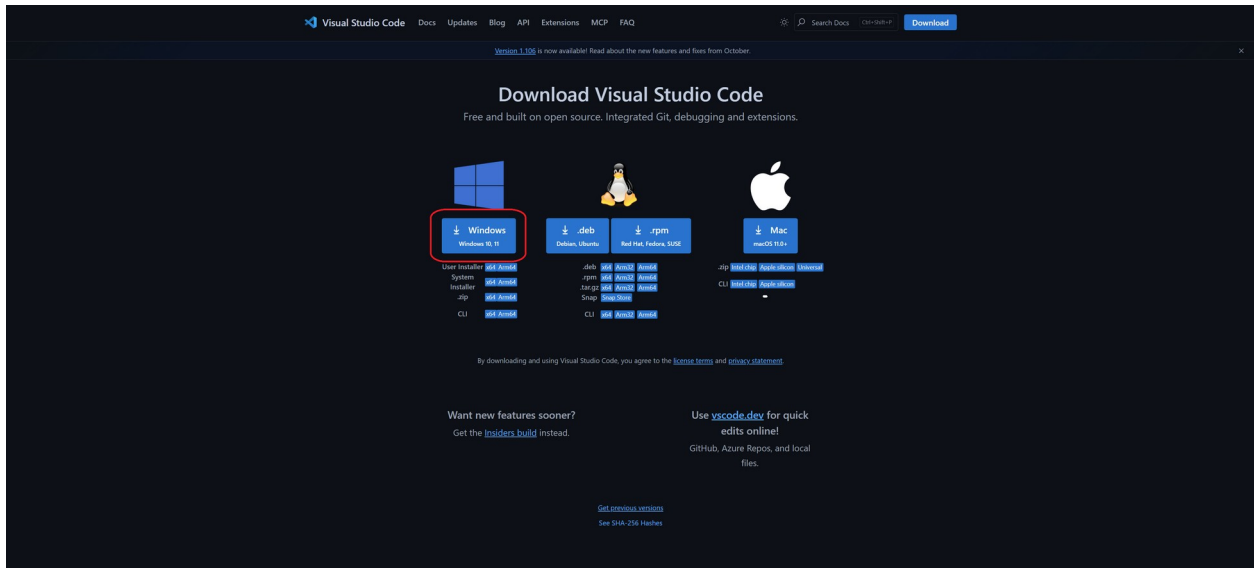
go to localhost:3000 this will bring you to open webUI. Create an account and then login into that account.



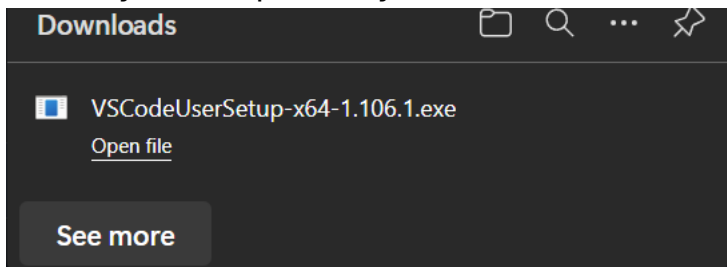
you can now chat with your AI that you installed. If you want to locally connect to the AI from another device, go to command prompt and enter the command ipconfig. Then from your other device go to a browser and in the URL then enter your IP address:3000. You should be able to use the AI that is hosted on your first device. Make sure that you are on the same network on both devices.

If you don't want to run the local AI on Open WebUI there is another way to run it with a python script. This method relies on using the Python script

along with html to make the UI and run it locally. To run the Python script, you first need to install VS code. Go to: [Download Visual Studio Code - Mac, Linux, Windows](#) and choose your OS.



Click on you OS (probably windows) and install VS code



Click on the download

Setup - Microsoft Visual Studio Code (User)

License Agreement

Please read the following important information before continuing.

Please read the following License Agreement. You must accept the terms of this agreement before continuing with the installation.

This license applies to the Visual Studio Code product. Source Code for Visual Studio Code is available at <https://github.com/Microsoft/vscode> under the MIT license agreement at <https://github.com/microsoft/vscode/blob/main/LICENSE.txt>. Additional license information can be found in our FAQ at <https://code.visualstudio.com/docs/supporting/faq>.

MICROSOFT SOFTWARE LICENSE TERMS

MICROSOFT VISUAL STUDIO CODE

These license terms are an agreement between you and Microsoft Corporation (or

☒ I accept the agreement

☐ I do not accept the agreement

Next Cancel

Check the box that says: "I accept the agreement" and click next

Setup - Microsoft Visual Studio Code (User)

Select Additional Tasks

Which additional tasks should be performed?

Select the additional tasks you would like Setup to perform while installing Visual Studio Code, then click Next.

Additional icons:

☐ Create a desktop icon

Other:

☐ Add "Open with Code" action to Windows Explorer file context menu

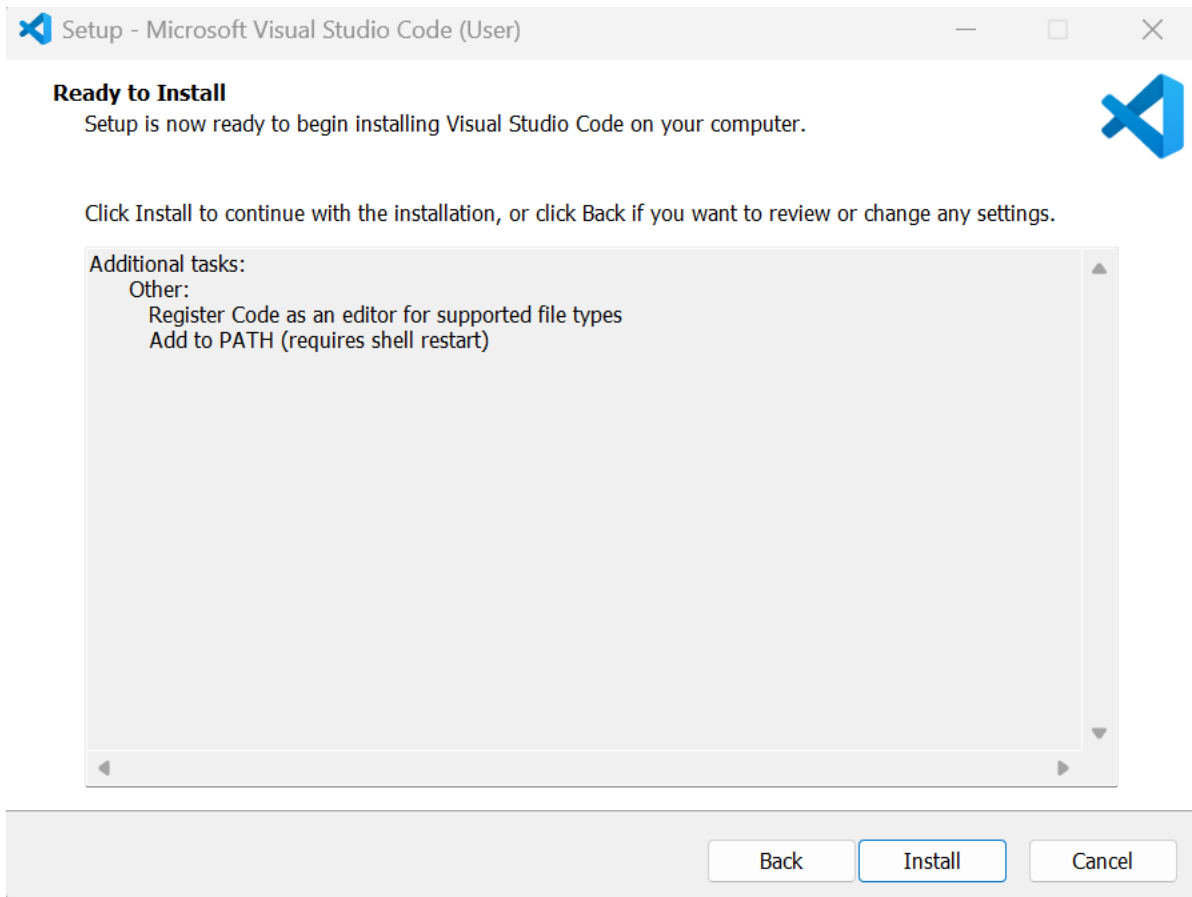
☐ Add "Open with Code" action to Windows Explorer directory context menu

☒ Register Code as an editor for supported file types

☒ Add to PATH (requires shell restart)

Back Next Cancel

Check the boxes that you want to check. If you don't know, just check these ones.



Then click install and wait for it to install.

Completing the Visual Studio Code Setup Wizard

Setup has finished installing Visual Studio Code on your computer. The application may be launched by selecting the installed shortcuts.

Click Finish to exit Setup.

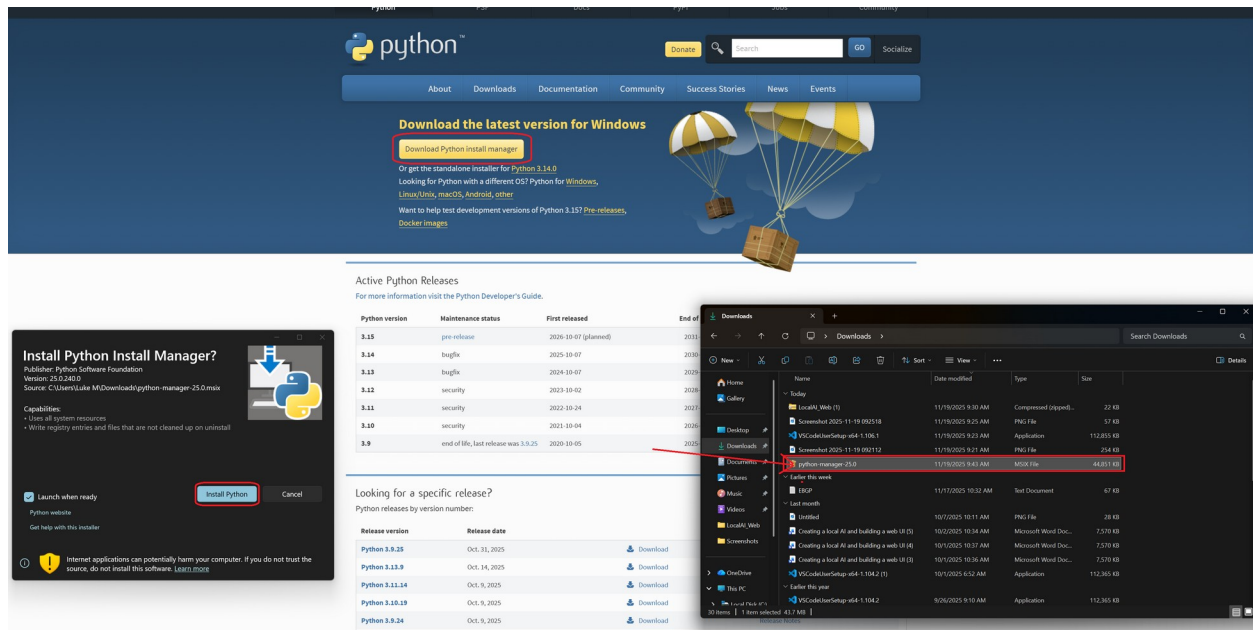


☒ Launch Visual Studio Code

Finish

Then click finish

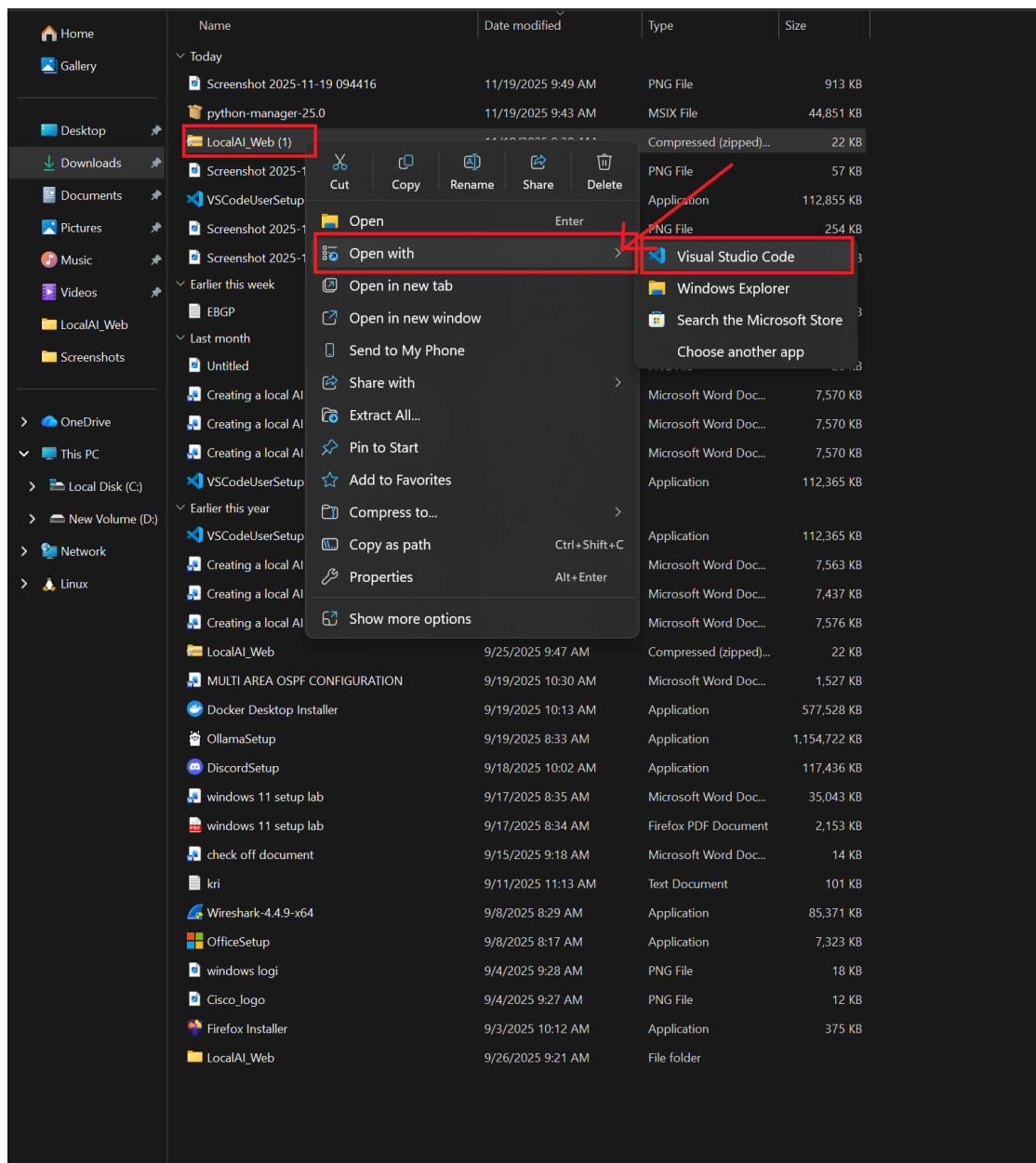
Now we must install python. First go to: [Download Python | Python.org](https://www.python.org/downloads/)



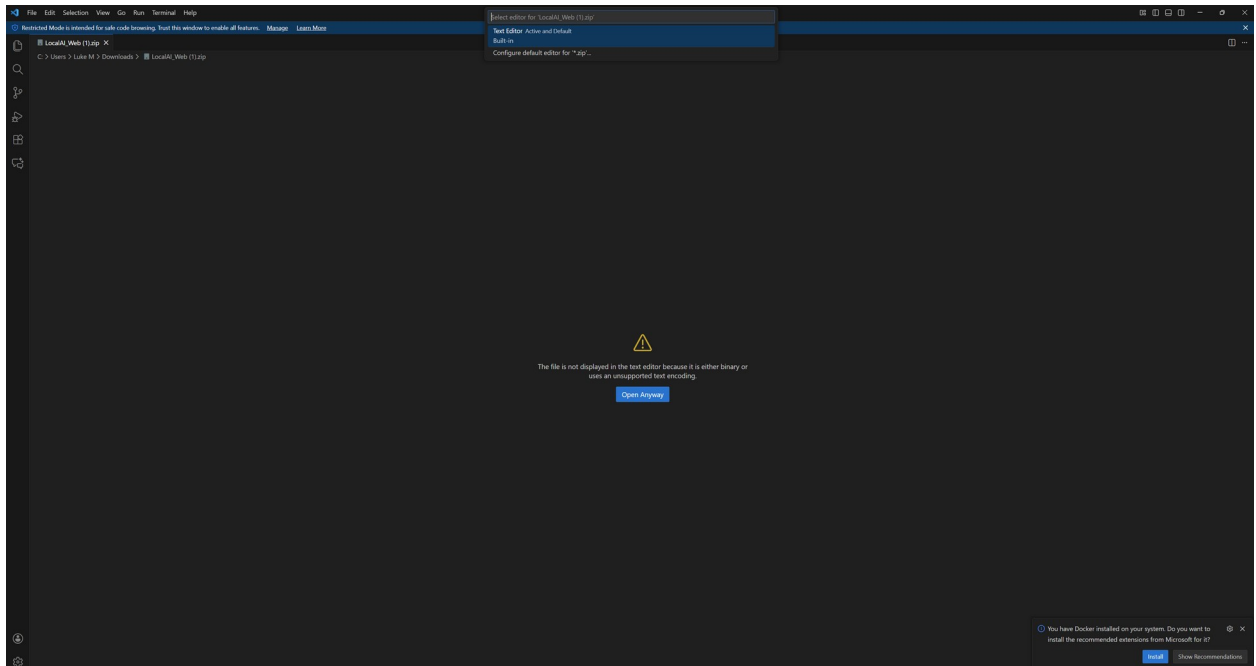
Download the install manager and then open the app installer and click install python

If you want to use the code that we made install this: [LocalAI_Web.zip](#)

Go to file explorer on your computer and find the local AI python script

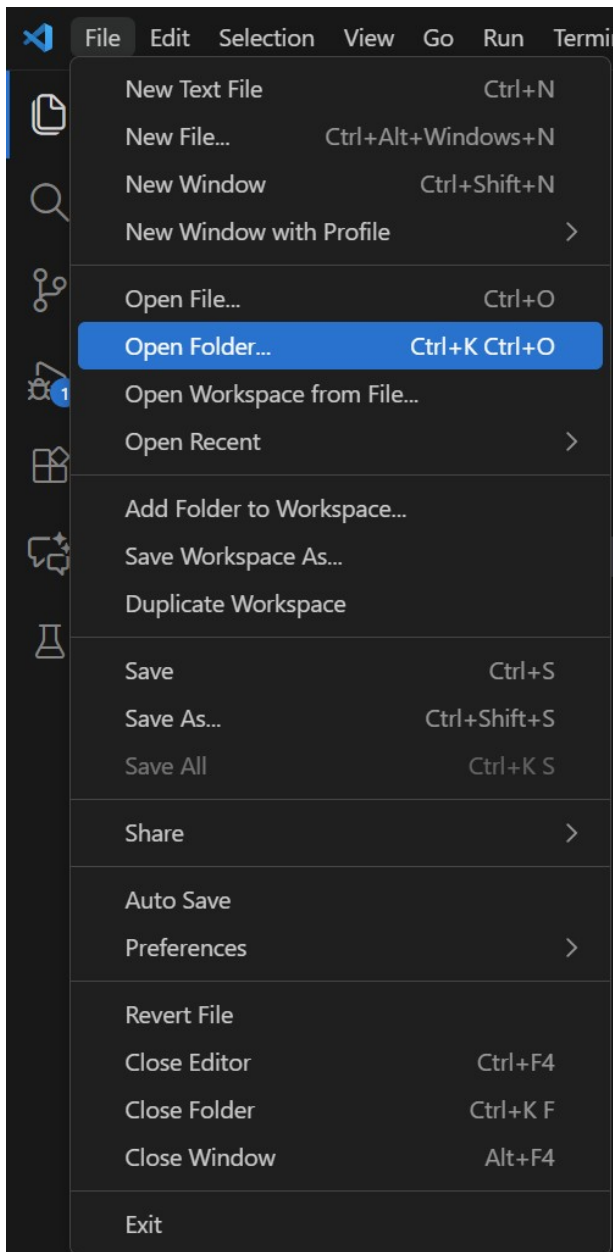


Right click on the zip file and choose open with then VS Code or choose the Choose another app and search for VS Cide if it is not in the list.

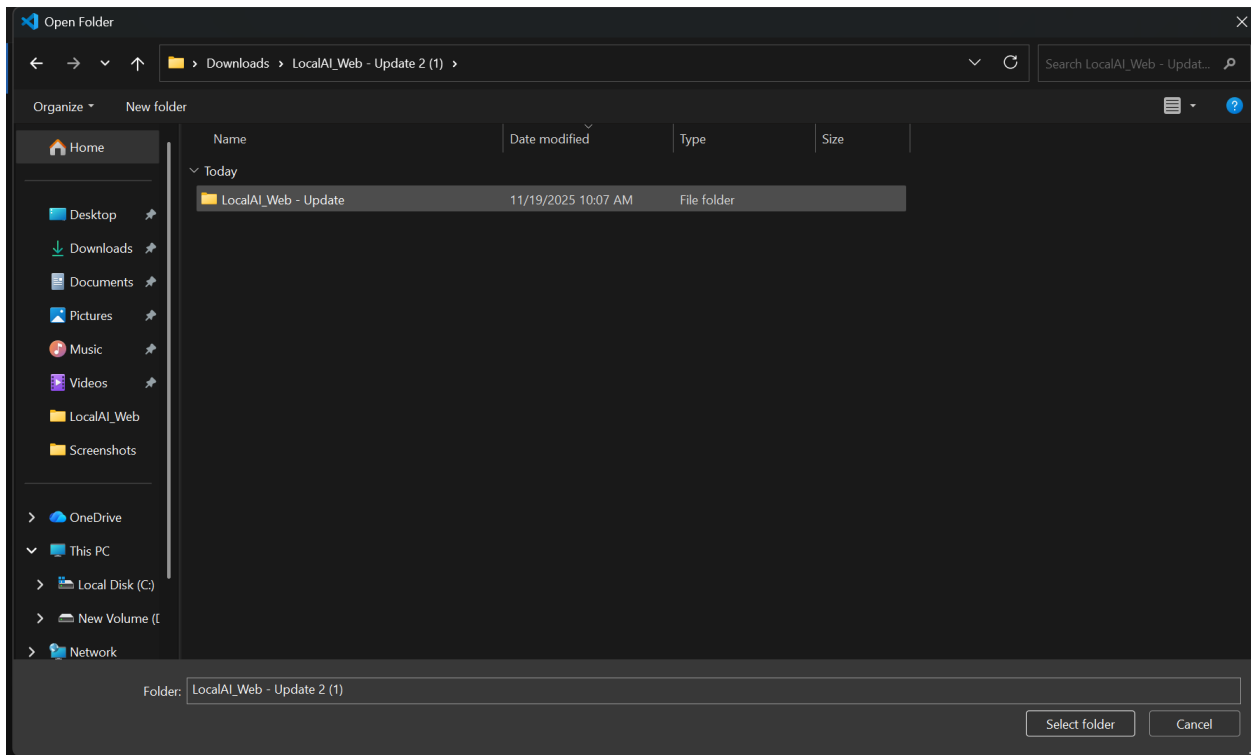


Select open anyway and then at the top select the text editor.

If the file does not work, find the script and extract it. Close the script in VS Code



Click file and open folder



Find the file and double click on it when the file should look something like this.

Select the file and then click select folder.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\server.py", line 10, in <module>
    from analysis import analyze_prompt
File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\analysis.py", line 5, in <module>
    from web_scraper import fetch_html
File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\web_scraper.py", line 1, in <module>
    import requests
ModuleNotFoundError: No module named 'requests'
PS C:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update>
PS C:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update>
PS C:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update> pip install requests
Collecting requests
  Downloading requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
Collecting charset_normalizer<4,>=2 (from requests)
```

Enter pip install request into the terminal

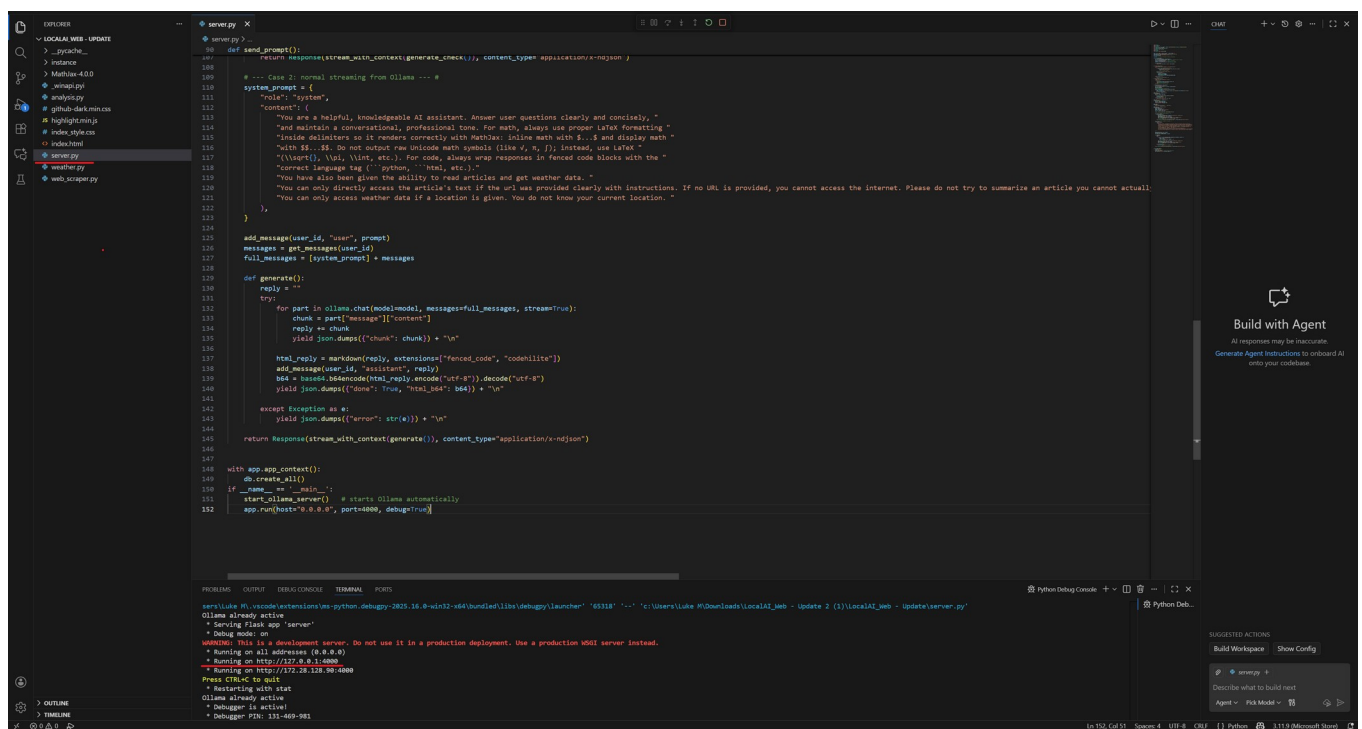
Do control + f5 and if it works then you should be able to run the script.

```

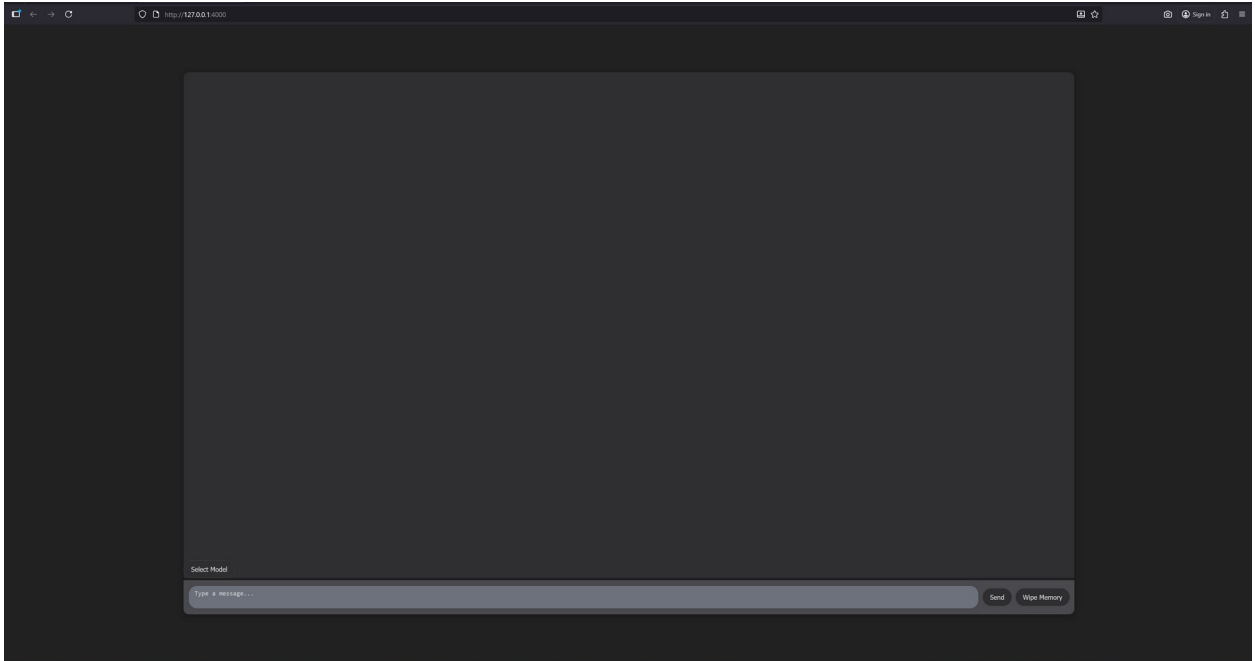
sers\Luke M\vscode\extensions\ms-python.debugpy-2025.16.0-win32-x64\bundled\libs\debugpy\launcher' '65299' '--' 'c:\Users\Lu
Traceback (most recent call last):
  File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\server.py", line 10, in <module>
    from analysis import analyze_prompt
  File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\analysis.py", line 5, in <module>
    from web_scraper import fetch_html
  File "c:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update\web_scraper.py", line 2, in <module>
    from bs4 import BeautifulSoup
ModuleNotFoundError: No module named 'bs4'
PS C:\Users\Luke M\Downloads\LocalAI_Web - Update 2 (1)\LocalAI_Web - Update> pip install bs4
Collecting bs4
  Downloading bs4-0.0.2-py2.py3-none-any.whl.metadata (411 bytes)
Collecting beautifulsoup4 (from bs4)

```

When I did control +f5 it gave me the no modules named bs4 so I did pip install bs4 if an error like this shows up install the module and the script should work.



Run the script with control + f5 and find the IP address underlined in the picture above.



Do control + click on the Ip address and it should show up with a UI like this above

Make sure you have the exact model of llama that is required, which is Llama3:8b. If you are not sure, then got to command prompt and type in ollama run llama3:8b

Problems:

When we were installing ollama we did not realize that you have to specify the number of parameters in the model that you install something like `ollama run llama3` will not be enough. We had to reinstall all of our models on both computers after realizing this. We then had an issue with the internet being very slow due to the amount of people using the bandwidth on the network, making the process take much longer. When using docker I was unable to find the command to enter the configuration of a container at first and had to remake the container every day for several days. We eventually found out that the `docker exec` command would enter the configuration. I also have a very limited knowledge of Linux terminal commands, and I took a very long time to install ollama in the container. I eventually learned that you don't even need to put ollama in the container, but I still found creative ways to get ollama working in a docker container. After Open WebUI was working and the ai was working on my computer, we were unable to get other people onto the AI through Open WebUI. We eventually found that some people were getting different IP addresses on the same network. We had to either statically set our IP addresses on our devices or find a device that was on the same network.

Conclusion:

We were able to learn many good skills in this lab like learning dockers capabilities with making containers that virtualize the operating system. We learned how ollama can run open-source AI engines locally on your computer. We then used Open WebUI to run ollama locally on the network allowing you to connect to the AI anywhere on the network.